



Odds Ratios Transcript

Hello everyone, welcome to lesson 1 video 4. In this video, we will discuss odds ratio, one of the most powerful ways of interpreting results from logit models. Let us interpret the results of odds ratio or the logistic regression form that we learnt $\log(p) = \beta_0 + \beta_1 x$. Let us interpret that $\log$ of odds or $\log$ of ratio of odds and interpret them. Now, what we do, let us say once we have got $\log$ of odds, we can actually transform it back into doing the same thing.

So, you have odds ratio as e^{β_1} because you have $\log$, you can take exponential and then it becomes e^{β_1} . Now, the resulting value e^{β_1} is called as the odds ratio because I told you that in the last video, if you take simply the derivative of this with respect to β_1 , you will get the rate of change of $\log$ of odds with respect to x , that is what is β_1 . So, this is what we are discussing here. Odds ratio is equals to e^{β_1} or exponential of β_1 , the resulting value is called as odds ratio. So, it tells us how the odds of a success are multiplied for one unit change in the predictor value because it is rate of change of $\log$ of odds.

So, just to help you understand what exactly it means, let us say we have a model for predicting the probability of someone passing the exam based on the hours of study that they have done. Now, given that the coefficient for hours is β_1 is equals to 0.5. Now, because β_1 is equals to 0.5, you can take e to the power 0.5 as 1.65. So, basically for any new additional hour of study that the student is making or is trying, the odds of passing increases by 65%. With every new hour of study that the student makes, their odds of passing increase by 0.65 or let us say they increase by 65%.

Now, think of if the value would have been $e^{-0.5}$. It means that with more and more hours of study, they are going to fail the exam because the odds of passing is going to be reduced by 65%. So, I was just telling you about what would happen if the coefficient value is negative. In that case, in this particular case, it is positive.

Hence, we are saying that odds of passing would increase by 65%. But, it could be negative as well. So, those are the cases when the increase of the variable causes the decrease in $\log$ of odds.

Now, what are those cases? This is one of the cases I was just telling you about. When odds ratio is greater than 1, basically your β_1 value is greater than 0. If β_1 is positive, it means that for every odds in change, there is a positive relationship. There is an increase which would happen.

If β_1 is less than 0, this is the second case which I was telling you about. In that case, if β_1 is less than 0, with one unit increase in x , there would be a decrease in the odds ratio. If β_1 is equal to 0, basically you are saying that the odds ratio is not at all dependent on the value of the variable itself.



Let us say in our loan example, if the value of the parameter would have been 0, it means that age is not at all a deterministic factor for finding out whether the individual would repay or not. I hope logistic regression has given you enough understanding of what exactly it is. It is a linear classification method.

That is what we call it in machine learning. But, it is mostly used in a lot of scenarios wherein your output variable is a categorical variable. What do I mean by categorical variable? A variable which has categories in it.

Categories as in yes or no, true or false, high, medium, low and so on. In all of these cases, we can use methods like logistic regression. That is what we have learned in this entire module.

I hope this has given you a good understanding of what logistic regression is. In summary, odds ratio provides a straightforward way to interpret coefficients from logic models. They tell us how predictor changes the likelihood of an outcome in relative terms.

With this, we complete our study of discrete response models.

-----END-----